Senzing

# Bootcamp

## Completion Recap

A record of everything you built and learned, module by module. Keep it, revisit it, and share it with your team.

| | |
|---|---|
| **BOOTCAMPER** | Bootcamper |
| **STARTED** | 2026-07-30 |
| **COMPLETED** | 2026-07-30 |
| **PROGRAMMING LANGUAGE** | Python |
| **PATH** | Core |
| **PLUGIN VERSION** | 0.6.1 |

### Modules in this recap

- Entity Resolution Concepts
- Discover the Business Problem
- SDK setup
- System verification
- Truth Set visualization
- Data collection
- Data Quality, Mapping, and Transformation
- Data processing
- Query, Visualize and Discover

## Run environment

**Operating system:** Ubuntu 24.04.4 LTS (x86_64)

**Python version:** 3.12.3

**Language runtime:** Python 3.12.3

**Senzing SDK:** 4.3.3

**Database:** SQLite

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# Entity Resolution Concepts

Completed July 30, 2026

## Information Shared

- What entity resolution is: deciding whether different records refer to the same real-world entity, then matching, relating, and deduplicating them.
- Two failure modes: false negatives (same entity split apart) and false positives (different entities merged).
- The conceptual pipeline: ingestion/standardization -> candidate selection (blocking) -> comparison/scoring -> classification -> entity clustering.
- Disclosed vs. discovered relationships (Senzing docs via MCP).
- What it produces: matched records grouped into unified entities, and relationships tracked between those entities.
- Senzing-specific: principle-based matching (vs. hand-written rules), pre-configured for people and organizations, differentiators (real-time, no model training, explainability with "why matched/why not/how" attribution, scalability as a composable library). Sourced from Senzing docs via the MCP server.

## Questions & Responses

- **Q:** Do you have any questions about entity resolution before we continue?
  - **R:** No.
- **Q:** Would you like a few quick questions to help the concepts stick?
  - **R:** Yes.
  - Q1 (principles vs. rules): answered correctly.
  - Q2 (disclosed relationship example): answered correctly.
  - Q3 (false negative definition): answered incorrectly (chose "possible match"); corrected and re-taught.
  - Q4 (explainability capability): answered correctly.
- **Q:** Are you ready to move on to the next module: Discover the Business Problem?
  - **R:** Yes.

## Actions Taken

- Presented the Entity Resolution Concepts primer and an optional 4-question knowledge check.
- No project files are created by this preamble module.

## End-of-Module Summary

**What you accomplished:**

- Learned what entity resolution is, its two failure modes, and the conceptual pipeline
- Learned how Senzing's principle-based matching differs from rule-based approaches
- Reinforced the concepts with a short knowledge check (3 of 4 correct, one concept re-taught)

**Files produced:**

- (no files - conceptual primer)

**Why it matters:**

These concepts are the vocabulary for the rest of the bootcamp - every later decision about matching, mapping, and resolution assumes them.

# Discover the Business Problem

Completed July 30, 2026

## Information Shared

- Data privacy reminder before working with any data.
- A pattern gallery: Customer 360, Fraud Detection, Compliance/KYC, and Vendor/MDM, each with problem, goal, sources, and business value (via Senzing docs).
- The generated Customer 360 scenario: Enformion and Equifax (real Senzing CORD data) as the two data sources, and how Senzing's principle-based matching resolves records across them without hand-written rules.
- How the module maps onto the rest of the bootcamp (SDK setup -> Data collection -> Data Quality, Mapping, and Transformation -> Data processing -> Query, Visualize and Discover).

## Questions & Responses

- **Q:** Would you like to see examples of common business problems that entity resolution can solve?
  - **R:** Yes.
- **Q:** Do any of these patterns match your situation?
  - **R:** Customer 360.
- **Q:** How would you like to define the business problem?
  - **R:** Generate a scenario (Business Case Offer accepted).
- **Q:** Does that summary capture your situation accurately?
  - **R:** Yes.
- **Q:** Will your entity-resolution results need to interface with other software?
  - **R:** No.
- **Q:** Where do you plan to deploy the final solution?
  - **R:** Local / on-premises.
- **Q:** Does this accurately capture your problem and approach?
  - **R:** Yes.

## Actions Taken

- Generated a Customer 360 business scenario backed by real Senzing CORD data (Enformion, Equifax).
- Created docs/data_architecture.md (data architecture and data flow diagrams).
- Created docs/business_problem.md (full business problem statement).
- Created config/data_sources.yaml (Enformion and Equifax source registry entries).
- Created README.md (project overview and business problem summary).
- Created docs/stakeholder_summary_module1.md.
- Recorded integration_targets: [] and deployment_target: local in config/bootcamp_preferences.yaml.

## End-of-Module Summary

**What you accomplished:**

- Defined a Customer 360 business problem grounded in real Senzing CORD data (Enformion, Equifax)
- Identified data sources, key matching criteria, and success criteria
- Captured integration and deployment preferences for later modules

**Files produced:**

- docs/business_problem.md - the full business problem statement
- docs/data_architecture.md - data architecture and data flow diagrams
- config/data_sources.yaml - registered data sources for the scenario
- README.md - project overview and business problem summary
- docs/stakeholder_summary_module1.md - stakeholder-facing summary

**Why it matters:**

This is the roadmap for the rest of the bootcamp - every later module (mapping, loading, querying) works against this defined problem and these named data sources.

# SDK setup

Completed July 30, 2026

## Information Shared

- The Senzing SDK was already installed (version 4.3.3, build 4.3.3.26191) via apt, so the install and EULA steps were skipped.
- A significant environment issue: PyPI senzing (4.1.2) and senzing_core (1.0.3) packages were installed in ~/.local/lib/python3.12/site-packages and were shadowing the SDK-shipped packages. The Senzing MCP server flags pip install senzing as an error-severity anti-pattern: the real packages ship with senzingsdk-runtime at /opt/senzing/er/sdk/python, and the PyPI ones are for unsupported community projects only.
- Two environment variables are required on this platform (from MCP sdk_guide install guidance): PYTHONPATH=/opt/senzing/er/sdk/python and LD_LIBRARY_PATH=/opt/senzing/er/lib. Without the latter, the import fails with "libSz.so: cannot open shared object file".
- The built-in evaluation license limits ingestion to 500 records; record 501 fails with SENZ9000.
- The SQLite database file is not auto-created - the schema DDL must be applied first.
- A freshly schema-created datastore has no registered Senzing configuration; without seeding one, data-source registration later fails with SENZ7221.
- Why a version query is not a valid engine test: the libraries and their support data can be present independently, so a wrong SUPPORTPATH satisfies a version probe and fails later at the first real engine call.

## Questions & Responses

- **Q:** Would you like to switch to /model opus + /effort high for this module?
  - **R:** Yes (set Opus 5.5 at high effort).
- **Q:** Are you done modifying the model and effort?
  - **R:** Yes.
- **Q:** Which database would you like to use?
  - **R:** SQLite.

## Actions Taken

- Detected the existing SDK install via both filesystem sentinels (/opt/senzing/er/lib/libSz.so, /opt/senzing/er/szBuildVersion.json) and dpkg; skipped install and EULA steps.
- Diagnosed and resolved the PyPI package shadowing by prepending the SDK path to PYTHONPATH.
- Created src/scripts/verify_sdk.py and confirmed the SDK reports version 4.3.3 (engine schema 4.3).
- Created the full project directory structure (src/scripts, src/resources, data/raw, data/mapping, data/temp, docs/progress, docs/visualizations, config, database, licenses).
- Applied the Senzing SQLite schema DDL to database/G2C.db (16 tables).
- Created config/engine_config.json from the MCP-returned JSON, overriding only the /tmp/sqlite default to

the project-relative database path.

- Created src/scripts/senzing-env.sh - a project-local environment script (no global shell config was modified), with bash/zsh self-location, a fail-loudly project-root guard, and a refusal to export an empty configuration.
- Created and ran src/scripts/seed_default_config.py, registering default config ID 3798842070.
- Created and ran src/scripts/test_connection.py, which exercises real engine-class calls: SzEngine.prime_engine() and SzDiagnostic.get_repository_info().
- Verified the env script's fail-loudly guard by sourcing it from a wrong root: it named the resolved root and returned 1 without closing the shell.
- Recorded database_type: sqlite in config/bootcamp_preferences.yaml.

## End-of-Module Summary

**What you accomplished:**

- Confirmed a working Senzing SDK 4.3.3 install and verified it from Python
- Found and fixed a real environment fault where unsupported PyPI packages shadowed the SDK-shipped ones
- Built a SQLite datastore with the Senzing schema and seeded its default configuration
- Proved the engine actually initializes with real engine-class calls, not just a version query

**Files produced:**

- config/engine_config.json - the Senzing engine configuration (MCP-sourced, project-relative DB path)
- src/scripts/senzing-env.sh - project-local environment script; source it before running Senzing code
- src/scripts/verify_sdk.py - SDK install/version check
- src/scripts/seed_default_config.py - registers the default Senzing configuration
- src/scripts/test_connection.py - engine and datastore connection test
- database/G2C.db - the SQLite datastore with the Senzing schema applied

**Why it matters:**

Every later module runs against this install and datastore, so getting the environment genuinely correct here - rather than appearing to work off shadowed packages - is what keeps the mapping and loading modules from failing in confusing ways.

# System verification

Completed July 30, 2026

## Information Shared

- System Verification uses synthetic records designed to resolve deterministically, never the Senzing Truth Set (that belongs to the separate Truth Set visualization module).
- A version query alone does not prove the engine works - the libraries and their support data can be present independently, so this module exercises real engine-class calls (SzEngine.prime_engine(), SzDiagnostic.get_repository_info()) before anything else.
- A freshly schema-created datastore has no registered data sources; the VERIFY source had to be registered before loading, or the load would fail with SENZ2207.
- The correct loading snippet reads records from an input file line by line; the MCP server's own anti-pattern list flags the hardcoded-records alternative as an error.
- Database operations (write count, read by entity ID, search by attributes) are all verified through generated SDK code - never direct SQL against database/G2C.db.

## Questions & Responses

- **Q:** Would you like to switch to /model sonnet for this module?
  - **R:** Yes (step down from Opus 5.5; effort stayed at high).
- **Q:** Are you done modifying the model and effort?
  - **R:** Yes.

## Actions Taken

- Designed 4 synthetic VERIFY records: a 3-record merge cluster ("Alex Quinn Verify", same DOB/address with trivial variation) and 1 distractor singleton ("Jordan Sample").
- Ran the System Verification checks: the installation checks - MCP connectivity, engine initialization, SDK initialization, code generation, build/compile, data source registration, data loading, and database operations - all passed, and results validation, reported separately, matched the prediction.
- Confirmed the merge cluster resolved to one entity (entity ID 1) and the distractor stayed its own entity (entity ID 4) - exactly as designed.
- Verified write count, read-by-entity-ID, and search-by-attributes all through generated SDK code.
- Purged all 4 synthetic VERIFY records via SzEngine.delete_record (not a full datastore purge, to preserve other state) and confirmed zero remain.
- Created src/system_verification/ with verification_data.jsonl, verify_init.py, verify_pipeline.py, register_data_sources.py, validate_results.py, verify_database_operations.py, and cleanup_verification_data.py - retained for reference.

## End-of-Module Summary

**What you accomplished:**

- Proved your entire Senzing setup works end to end: initialization, code generation, compilation, loading,

entity resolution, and database operations

- Watched Senzing correctly merge near-duplicate records into one entity while keeping an unrelated record separate - the core behavior the rest of the bootcamp depends on
- Confirmed the environment is clean again (synthetic data purged) before real project data arrives

**Files produced:**

- src/system_verification/verification_data.jsonl - the 4 synthetic test records
- src/system_verification/verify_init.py - SDK initialization check
- src/system_verification/verify_pipeline.py - file-based record loader
- src/system_verification/register_data_sources.py - idempotent data-source registration
- src/system_verification/validate_results.py - deterministic entity-resolution validation
- src/system_verification/verify_database_operations.py - write/read/search verification
- src/system_verification/cleanup_verification_data.py - synthetic-data purge

**Why it matters:**

This is the proof that Senzing is actually resolving entities correctly on your machine, with your install, before you load anything that matters.

# Truth Set visualization

Completed July 30, 2026

## Information Shared

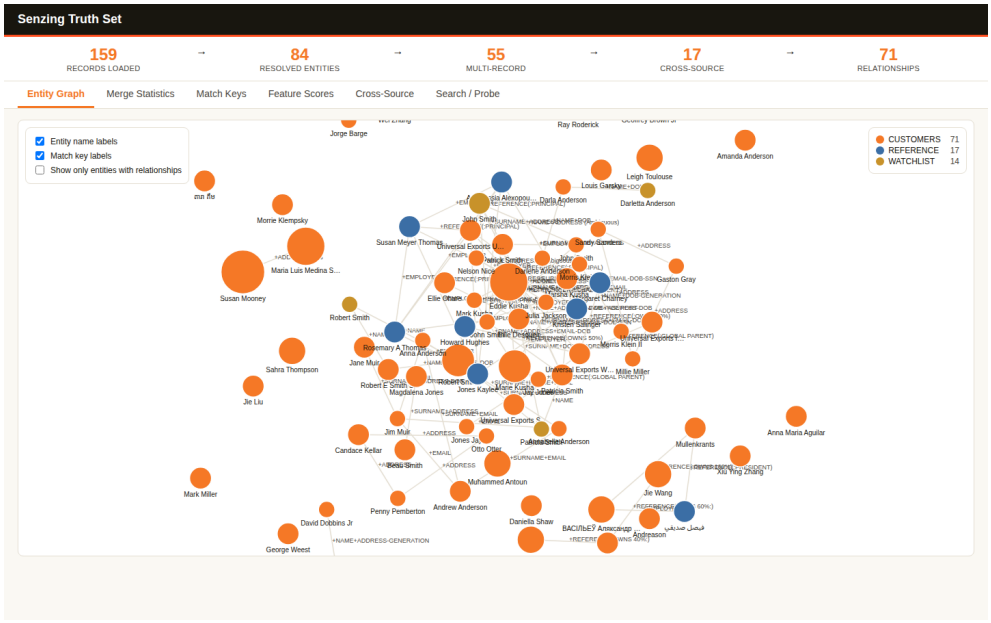
- The Senzing Truth Set is the demo dataset with a known ground truth: 159 records across CUSTOMERS (120), REFERENCE (22), and WATCHLIST (17), retrieved via the MCP server (provenance mcp_primary).
- Senzing resolved those 159 records into 84 entities - 55 multi-record (duplicates found), 17 spanning more than one data source, with 71 relationships discovered.
- Match keys name the features that drove each resolution: +NAME+DOB+ADDRESS led with 12 resolutions out of 28 distinct keys.
- Searching "Robert Smith" returns three *distinct* resolved entities rather than one merged blob - the point being that Senzing merges duplicates while keeping genuinely different people apart.
- The saved snapshot renders every tab offline from embedded data, but the live Why?, How?, and search actions need the running engine.
- The visualization renders fully offline - D3 v7 is inlined, with zero external fetches.

## Questions & Responses

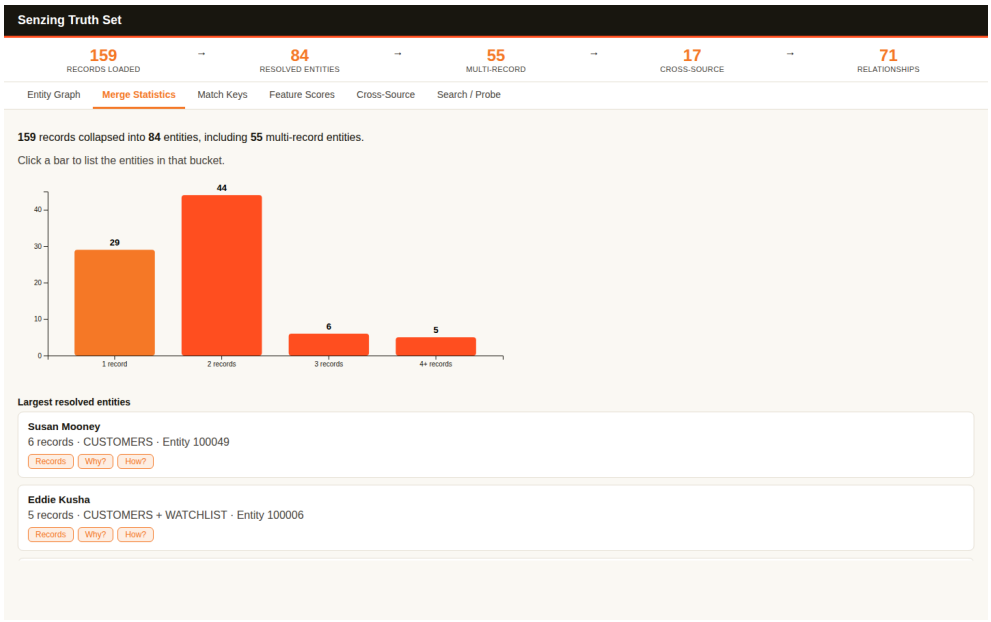
- **Q:** Would you like to switch to /model opus for this module?
  - **R:** Yes (effort already at high).
- **Q:** Are you done modifying the model and effort?
  - **R:** Yes.
- **Q:** Are you ready to continue? (after the guided tour)
  - **R:** Yes.
- **Q:** Ready for me to stop the visualization server and clean up the Truth Set data?
  - **R:** Yes.

## Actions Taken

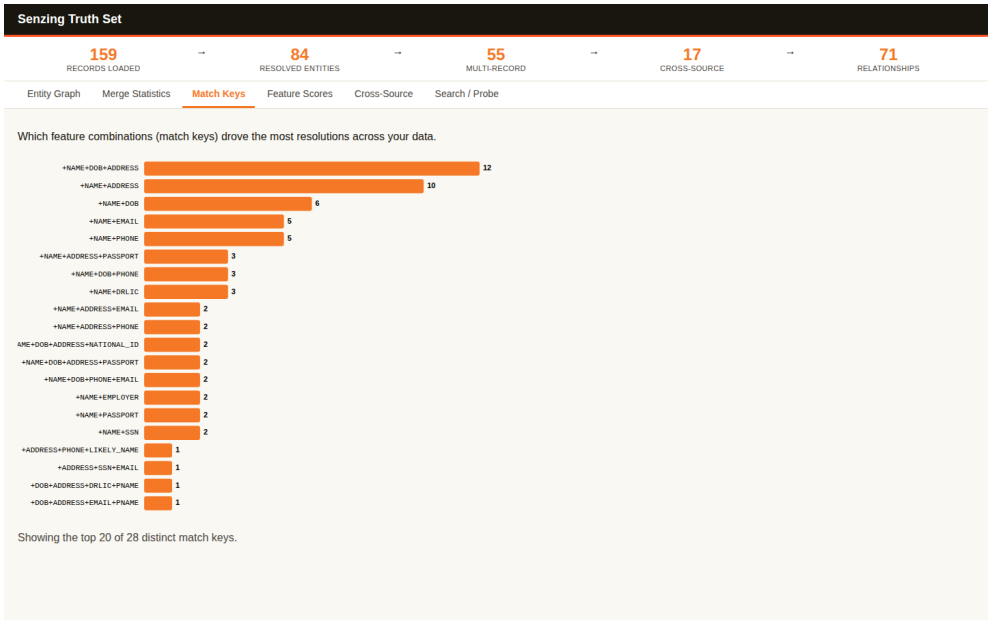
- Acquired the Truth Set via the MCP server and fetched all 159 records; verified per-source counts matched exactly (120/22/17) with zero unparseable lines.
- Registered CUSTOMERS, REFERENCE, and WATCHLIST from the codes actually present in the data, then loaded all 159 records with 0 errors.
- Built the standalone snapshot first (the retained artifact), then started the live app on port 8080.
- Verified all 10 API endpoints against the contract. Two initially appeared to fail; inspecting the payloads showed WHY_RESULTS and HOW_RESULTS are nested under result - the check was wrong, not the server, and both carry real data (WHY_KEY: +NAME+DOB, rule SNAME_SSTAB).
- Confirmed the snapshot's six tabs match the running server's exactly, so the keepsake is not stale.
- Captured and visually verified one screenshot per tab from the live server.
- Stopped the web service, confirmed port 8080 released, then purged all 159 Truth Set records and confirmed zero remain.



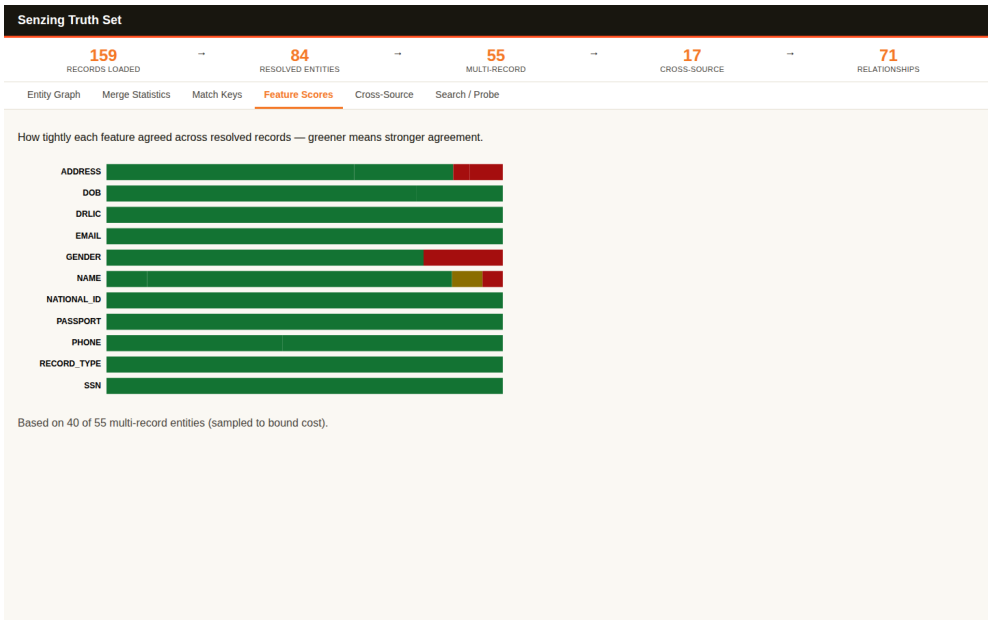

*Entity Graph - 84 resolved entities spread across the force layout, colored by data source (CUSTOMERS 71, REFERENCE 17, WATCHLIST 14) and sized by record count, with match-key edge labels*



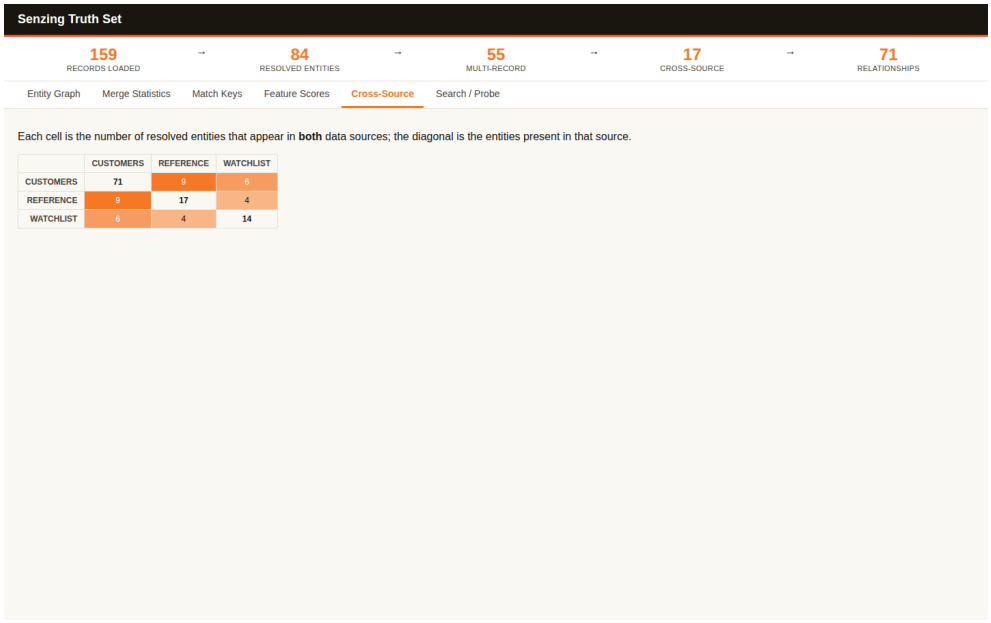

*Merge Statistics - records-per-entity histogram (29 singletons, 44 two-record, 6 three-record, 5 four-plus) above the largest resolved entities, Susan Mooney at 6 records and Eddie Kusha at 5 spanning CUSTOMERS + WATCHLIST*



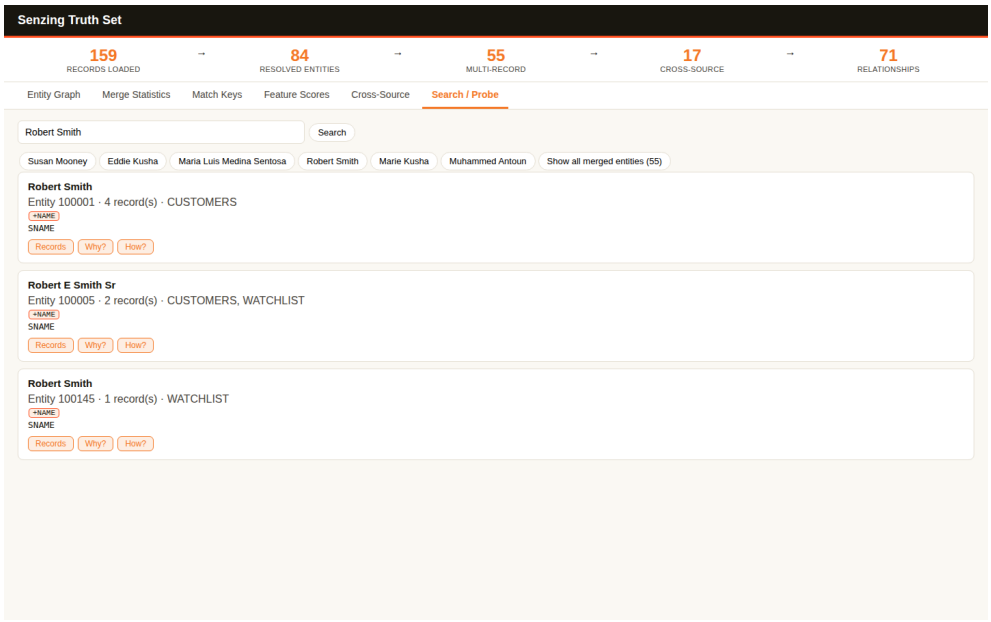

*Match Keys - +NAME+DOB+ADDRESS leads at 12 resolutions, followed by +NAME+ADDRESS at 10, from 28 distinct match keys*




*Feature Scores - per-feature agreement across resolved records, sampled from 40 of 55 multi-record entities; DRLIC, EMAIL, PASSPORT and SSN agree fully while GENDER and NAME show partial disagreement*




*Cross-Source - 3x3 overlap matrix: CUSTOMERS and REFERENCE share 9 entities, CUSTOMERS and WATCHLIST 6, REFERENCE and WATCHLIST 4*




*Search / Probe - the query "Robert Smith" returning three distinct resolved entities, including Robert E Smith Sr spanning CUSTOMERS and WATCHLIST, each with Records / Why? / How? actions*

## End-of-Module Summary

**What you accomplished:**

- Watched the real Senzing Truth Set resolve in an interactive web app - 159 records collapsing into 84 entities, live on your own machine
- Explored the resolutions from six angles: the entity graph, merge statistics, match keys, feature scores, cross-source overlap, and search
- Saw the distinction that matters - three different Robert Smiths kept apart while their duplicates merged
- Kept a permanent offline copy of the whole app plus six verified screenshots

**Files produced:**

- docs/visualizations/truthset_verification.html - the standalone, fully offline snapshot of the app (411 KB)

- docs/visualizations/truthset_verification-*.png - six screenshots, one per tab
- src/system_verification/truthset_data.jsonl - the 159 Truth Set records
- src/system_verification/register_truthset_sources.py - registers the Truth Set data sources
- src/system_verification/load_truthset.py - loads the Truth Set
- src/system_verification/cleanup_truthset.py - purges the Truth Set records

**Why it matters:**

This was practice data, built to contain matches - concrete proof of what entity resolution does, on a dataset with known ground truth, before you point Senzing at your own messier data. The saved snapshot is yours to share with your team.

# Data collection

Completed July 30, 2026

## Information Shared

- Random sampling from two sources destroys cross-source overlap: an initial 3,000-record preview batch from each of Enformion and Equifax shared only 15 distinct names out of ~2,800 - confirming the bootcamp's own warning about naive sampling.
- The built-in evaluation license caps ingestion at 500 records; after expanding to 5,000 total records, the collected total genuinely exceeded that limit, triggering the License Key gate.
- Senzing is locally hosted - no data flows to Senzing, Inc. - which is why the raw CORD files are still excluded from version control as a matter of general data-handling hygiene.

## Questions & Responses

- {initial pass: none - both sources already had provenance: cord recorded in Module 1, so the "how would you like to provide the data" question was correctly skipped per INV-006}
- **Q:** (Bootcamper request) I'd like to have 5000 records.
  - **R:** Reopened Data collection; expanded to 2,500 records per source (5,000 total).
- **Q:** Which best describes your Senzing License Key situation?
  - **R:** No - request a free evaluation license now through the bootcamp.
- **Q:** What's your first name (for the license request)?
  - **R:** Bootcamper.
- **Q:** What's your last name?
  - **R:** Example.
- **Q:** What's your work email address?
  - **R:** bootcamper@example.com.
- **Q:** How did you hear about Senzing?
  - **R:** Senzing bootcamp.
- **Q:** Send this evaluation-license request, including your name and work email, to Senzing?
  - **R:** Yes, send it.

## Actions Taken

- Initial pass: downloaded 3,000-record preview batches of Enformion and Equifax to compute cross-source name overlap (15 shared names found); built 250-record samples per source.
- On request, expanded scope: downloaded a capped preview batch from each source (as many records as the download endpoint serves per request) and found 98 shared name-keys (115 Enformion records, 103 Equifax records).
- Rebuilt final samples at 2,500 records per source (5,000 total), keeping all 98 matched pairs whole and filling the remainder randomly (seed 42) for singletons and non-matches.
- Re-validated both files at the new scale: readable, non-empty, valid UTF-8 JSONL, expected record count, consistent DATA_SOURCE, RECORD_ID present on every record - both passed.

- Updated config/data_sources.yaml and config/cord_metadata.yaml (new SHA-256 hashes) to reflect the expanded samples.
- Ran the License Key gate: 5,000 records exceeds the 500-record evaluation limit, so submitted an in-flow evaluation-license request via the Senzing MCP server (submit_feedback, category license_request) after explicit bootcamper consent to send name and work email - continuing on the built-in evaluation license while the emailed key is pending.
- Created .gitignore excluding data/raw/, database/, and licenses/ before any commit.
- Ran the SQLite load-time check: no MCP-confirmed timing threshold applies at this volume, so no warning was presented; Module 6 will use the threaded loading pattern automatically (record count > 500).

## End-of-Module Summary

**What you accomplished:**

- Collected real data for both sources in your Customer 360 scenario, expanded on request to 2,500 records each (5,000 total)
- Avoided a subtle, easy-to-miss failure: naive random sampling would have produced almost zero cross-source matches for Senzing to find - 98 real matched pairs were identified and preserved instead
- Requested a free Senzing evaluation license in-flow to cover the expanded volume, with explicit consent before any personal details were sent
- Confirmed both files are clean and ready for mapping, with a full audit trail of how and why they were sampled

**Files produced:**

- data/raw/enformion.jsonl - 2,500 Enformion records
- data/raw/equifax.jsonl - 2,500 Equifax records
- docs/data_source_locations.md - source locations and access methods
- docs/data_collection_checklist.md - inventory table and validation checklist
- docs/security_compliance.md - data privacy and handling notes
- config/cord_metadata.yaml - content-hash snapshot for change detection
- .gitignore - excludes raw data, the datastore, and licenses from version control

**Why it matters:**

Data Quality, Mapping, and Transformation needs real, validated files to work with - and because these two sources actually overlap on 98 real people/organizations, the rest of the bootcamp will have genuine cross-source matches to find, not an empty result set.

# Data Quality, Mapping, and Transformation

Completed July 30, 2026

## Information Shared

- The Entity Specification confirms two supported record shapes: the recommended FEATURES array and the still-supported legacy flat form ("In prior versions we allowed a flat JSON structure with a separate sub-list for each feature... While we still support that..."). Equifax uses both - FEATURES for persons, flat root attributes for organizations.
- get_record_preview asks Senzing how it actually reads a record without loading it, which is a stronger readiness test than reading attribute names against the specification by eye. It requires the record's DATA_SOURCE code to be registered first (SENZ2207 otherwise) even though it writes nothing.
- Senzing derives usage type from attribute-name prefixes: BUSINESS_ADDR_LINE1 becomes feature ADDRESS with USAGE_TYPE: BUSINESS, FAX_PHONE_NUMBER becomes PHONE with USAGE_TYPE: FAX.
- Completeness must be measured per record against the features applicable to that record's own RECORD_TYPE; averaging person-only features across an organization-heavy source penalizes data that could not exist.
- The quality score covers structural properties of one source in isolation. It says nothing about whether two sources resolve correctly against each other - that is established after loading, by the match-key audit in Data processing.

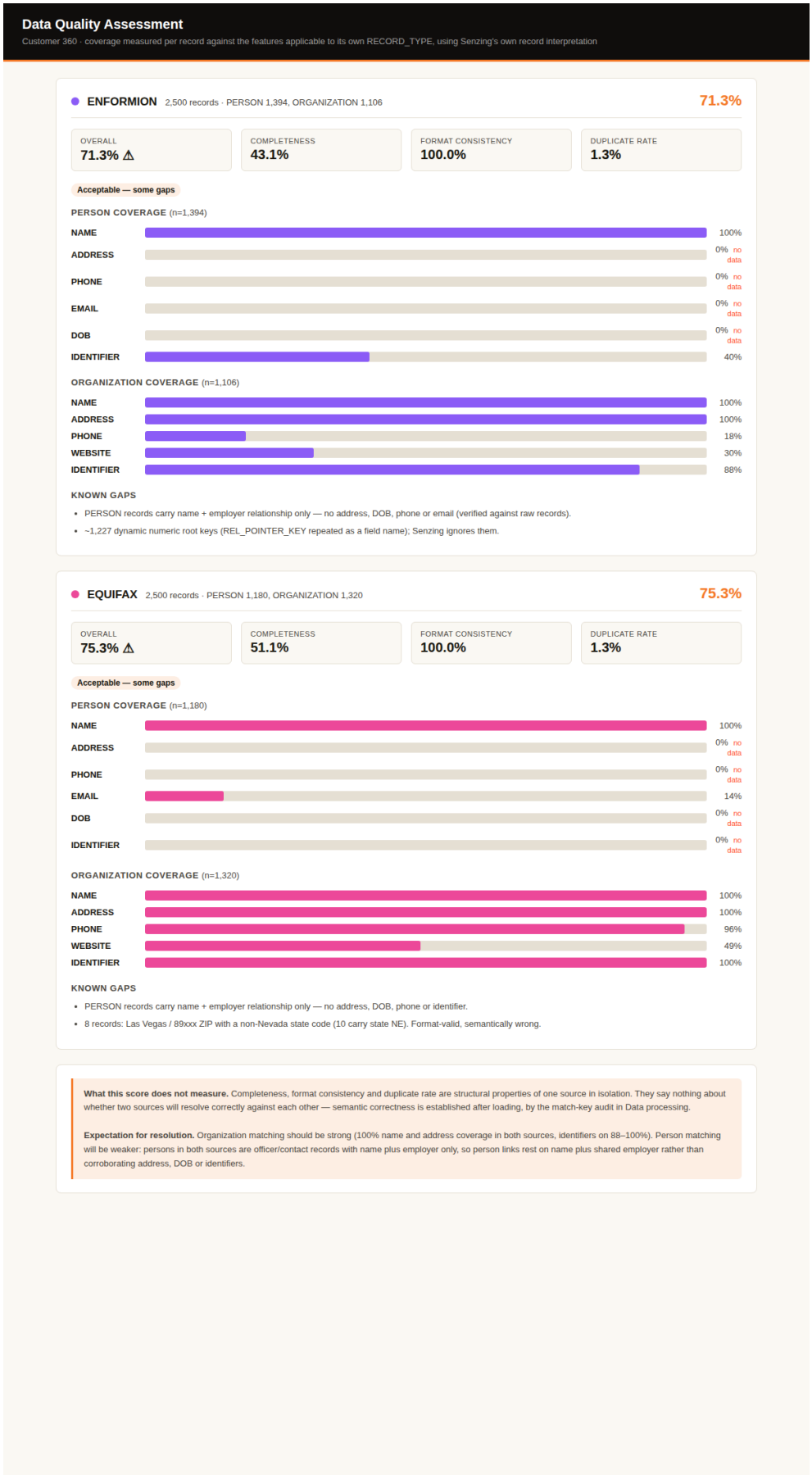
## Questions & Responses

- **Q:** Would you like to switch to /model opus + /effort high for this module?
  - **R:** Yes.
- **Q:** Are you done modifying the model and effort?
  - **R:** Yes.
- **Q:** Your CORD sources ENFORMION and EQUIFAX are already in Senzing-loadable form. Would you like to skip the mapping phase and proceed directly to loading?
  - **R:** Yes, skip.
- **Q:** Would you like a visual of the quality assessment?
  - **R:** Yes.

## Actions Taken

- Downloaded the Senzing Entity Specification to docs/reference/senzing_entity_specification.md (its saved size matched the response's size_bytes) and used it as the authoritative reference throughout.
- Profiled both sources' field structures, finding Enformion carries ~1,227 dynamic numeric root keys (the REL_POINTER_KEY repeated as a field name) atop ~23 stable fields.
- Registered ENFORMION and EQUIFAX data source codes (required before preview, and needed by loading anyway).
- Ran a readiness check on 200 records per source: structural check passed 200/200 for both, and get_record_preview confirmed Senzing extracts the intended features from every record.

- **Corrected an initial misreading:** Equifax's prefixed attribute names looked non-compliant against the specification, but the engine's own interpretation showed it derives usage types from the prefixes. Judging by eye would have sent good data through unnecessary remapping.
- Computed quality scores per record against each record's own RECORD_TYPE: ENFORMION 71.3%, EQUIFAX 75.3% - both in the "acceptable with gaps" band.
- Sanity-checked the 0% person-coverage figures against the raw records (0 of 1,394 Enformion persons carry any address attribute), confirming real data absence rather than a presence-test artifact.
- Found 8 Equifax records with a Las Vegas / 89xxx-ZIP address carrying a non-Nevada state code - format-valid but semantically wrong, and invisible to the format check.
- Fast-pathed both sources: mapping_status: complete, fast_pathed: true, file_path still pointing at the original data/raw/ files.
- Wrote docs/mapping/data_lineage.yaml and verified all five fast-path invariants for both entries (source == output, records_in == records_out, zero rejected, null script, fast_pathed true).
- Generated and verified docs/visualizations/data_quality_assessment.html - brand tokens applied, zero external fetches, all data-sourced strings escaped, source colors assigned from the data present. Opened the rendered page to confirm the bars drew and the band labels match the computed scores.




*Data Quality Assessment - per-RECORD_TYPE coverage bars for ENFORMION (71.3%) and EQUIFAX (75.3%), showing 100% NAME coverage for both types and 100% ADDRESS for organizations, with person ADDRESS/PHONE/DOB explicitly marked "no data"*

## End-of-Module Summary

### What you accomplished:

- Verified both sources are genuinely Senzing-loadable using the engine's own record interpretation, not a by-eye reading of field names
- Scored data quality correctly per record type, avoiding the averaging error that would have flagged clean organization data as needing remediation
- Skipped mapping entirely for both sources, with a full lineage record proving no transformation occurred
- Learned what the resolution will and won't be able to do: strong organization matching, weaker name-plus-employer person matching

**Files produced:**

- docs/reference/senzing_entity_specification.md - the authoritative Entity Specification
- docs/data_source_evaluation.md - the evaluation report for both sources
- docs/mapping/data_lineage.yaml - fast-path lineage entries, invariants verified
- docs/visualizations/data_quality_assessment.html - offline, brand-styled quality visual
- docs/visualizations/data_quality_assessment.png - rendered screenshot
- src/scripts/register_project_sources.py - registers ENFORMION and EQUIFAX
- src/scripts/preview_records.py - asks Senzing how it reads a record
- src/scripts/readiness_check.py - structural + preview-based readiness check
- src/scripts/quality_assessment.py - per-record-type quality scorer
- src/scripts/generate_quality_viz.py - the visualization generator

**Why it matters:**

Both sources go into loading unmodified, which means anything the resolution gets wrong is a real property of the data rather than something a mapping step introduced - and you now know in advance which matches to expect and which to be skeptical of.

# Data processing

Completed July 30, 2026

## Information Shared

- The loader's architecture follows the *production* volume target, not the bootcamp dataset: sdk_guide switches template at a 500-record cutover, and a medium-tier target selects the threaded pattern while labeling the single-threaded one "demo-only".
- Sorted or grouped input costs 2-10x throughput through lock contention, rated error severity by Senzing's own anti-pattern list. This project's files were grouped (matched pairs written first during collection), so shuffling before load genuinely mattered.
- Redo must be drained by looping on get_redo_record() returning empty. Using count_redo_records() as the loop sentinel is a documented anti-pattern: it table-scans per call, and because redo generates more redo, a count-driven loop is O(n squared).
- In a match key, + means the feature contributed and - means it detracted. A feature detracting on many cross-source comparisons can mean two source fields measuring different things were mapped to the same feature, silently suppressing legitimate merges.
- The Entity Specification's TRUSTED_ID rule: same TRUSTED_ID_TYPE with a different number forces records apart, but different types "do not interact at all" - the type namespaces the comparison.
- Whether an export carries RELATED_ENTITIES depends on the flag set, not the method, so one row must be dumped and its top-level keys read before writing any parser.

## Questions & Responses

- **Q:** In production - not in this bootcamp - how many records do you expect to load?
  - **R:** Option 3 - medium production (>500,000 up to 10,000,000).
- **Q:** Loading this data volume into SQLite may slow entity resolution as the database grows. How would you like to proceed?
  - **R:** Proceed on SQLite.

## Actions Taken

- Verified CORD data freshness by SHA-256 against the Module 4 snapshot (both sources unchanged) and backed up the database to data/backups/G2C.db.pre-load.
- Built src/load/load_senzing.py from the MCP threaded pattern, adding input shuffling, per-record error logging, throughput reporting, checkpoint/resume, and one shared factory.
- Built src/load/process_redo.py as a drain variant of the MCP continuous-daemon pattern, with the count used for monitoring only.
- Test-loaded 50 records, then loaded 2,500 Enformion (2.4s) and 2,500 Equifax (4.3s) - 5,000 total, zero errors.
- Drained the redo queue: backlog 443 to 0, processing 452 records (more than the backlog, because redo cascades).
- **Measured the active license rather than assuming it:**

recordLimit: 0 (no cap), Senzing Internal EVAL expiring 2027-03-12. This corrected the generic 500-record figure relayed in earlier modules, which never applied to this machine.

- Dumped one export row before parsing; confirmed RELATED_ENTITIES present with populated MATCH_KEY, and confirmed SZ_EXPORT_ALL_FLAGS is absent from the Python binding.
- Ran the ER statistics and match-key audit in a single export pass with relationships deduplicated by (min_id, max_id).
- Investigated the -TRUSTED_ID suppressor with concrete evidence (88.5% single-source, 11.3% cross-source) and cleared it: zero cross-source single-source-pair relationships carry it; all are same-type force-apart between different businesses sharing an address. Entity 200021 proved the namespacing works, holding both a BID and an EFX_ID after merging across sources.
- Verified 3 cross-source merges against raw source records (all true positives) and inspected the largest entity (400304, MIKE FIGLEY, 3 records) for over-matching - correct.
- Analyzed the under-matching side: 128 cross-source POSSIBLY_SAME pairs, 122 on +NAME alone, with evidence showing name is the only shared feature on those person pairs.

## End-of-Module Summary

**What you accomplished:**

- Loaded all 5,000 records across two sources with zero errors and drained the redo queue
- Built a take-home loader whose architecture matches your stated production volume, with every documented loading anti-pattern deliberately avoided
- Ran a match-key audit that caught a suspicious 88.5% suppressor and, with evidence, proved it was the engine working correctly rather than a mapping defect
- Validated the results against actual records instead of aggregate statistics: 3/3 merges true positives, zero false merges

**Files produced:**

- src/load/load_senzing.py - threaded production loader with shuffling, error logging, throughput reporting, checkpoint/resume
- src/load/process_redo.py - redo drain with monitoring-only counts
- src/load/register_data_sources.py - idempotent data-source registration
- src/scripts/er_statistics.py - single-pass ER statistics and match-key audit
- src/scripts/probe_export_row.py, probe_suppressor.py, probe_undermatching.py, probe_entity.py - evidence probes
- src/scripts/check_license.py - measures the active license
- docs/er_statistics.json, docs/results_validation.md, docs/uat_results.md, docs/loading_strategy.md, docs/stakeholder_summary_module6.md
- config/license.json - the measured license record

**Why it matters:**

The data is resolved and the results are trustworthy for a specific reason: the one signal that looked like a defect was chased down with evidence rather than accepted or dismissed, and the limits on person

matching are understood as a property of the source data rather than a mystery.

# Query, Visualize and Discover

Completed July 30, 2026

## Information Shared

**Headline entity:** AUGMENTI CONSULTING LLC (entity 200021), 2 records from 2 data sources, ENFORMION and EQUIFAX. No entity in your data spans more sources, none that spans both holds more records, and of the entities that tie with it, it has the lowest entity ID.

- ENFORMION record ENF-118734: AUGMENTI CONSULTING LLC, with its Enformion business ID (BID)
- EQUIFAX record EFX-55102: AUGMENTI CONSULTING LLC, with its Equifax ID (EFX_ID)
- **Why they resolved together:** match key +NAME+ADDRESS, the same name at the same address in two systems. Their business IDs are of different types, which do not interact, so nothing held the two records apart.

**Also shared:**

- Reading match keys in your own data's terms: + means the feature contributed, - means it detracted, and the difference between +NAME alone (queued for review) and +NAME+ADDRESS (resolved) is the whole gap between 20 merges and 128 review items.
- Flag defaults differ per method and this bites silently: SZ_ENTITY_DEFAULT_FLAGS (get_entity) carries both RECORDS[] and RECORD_SUMMARY[], but SZ_SEARCH_BY_ATTRIBUTES_ALL carries only the summary - so a habit learned on get_entity yields an empty records list on search, with a correct field name and no error.
- NAME_ORG is the organization name attribute; an organization name sent as NAME_FULL matches nothing and raises nothing. Roughly half this data is organizations, so the search tries both.
- Precision and recall require a truth set. Without one, only proxy indicators are available - entity size distribution, cross-source rates, singleton rate, and manual sampling.
- Why vs. How: Why explains the current decision between two records or entities; How shows one entity's chronological construction history. "Why are these two people in the same room?" versus "watch the security footage of everyone entering."
- Python's graph methods take native collections (entity_ids: List[int]), not the {"ENTITIES":[...]} document the Java/C# signatures imply - and network link endpoints are MIN_ENTITY_ID/MAX_ENTITY_ID, not ENTITY_ID/RELATED_ENTITY_ID.

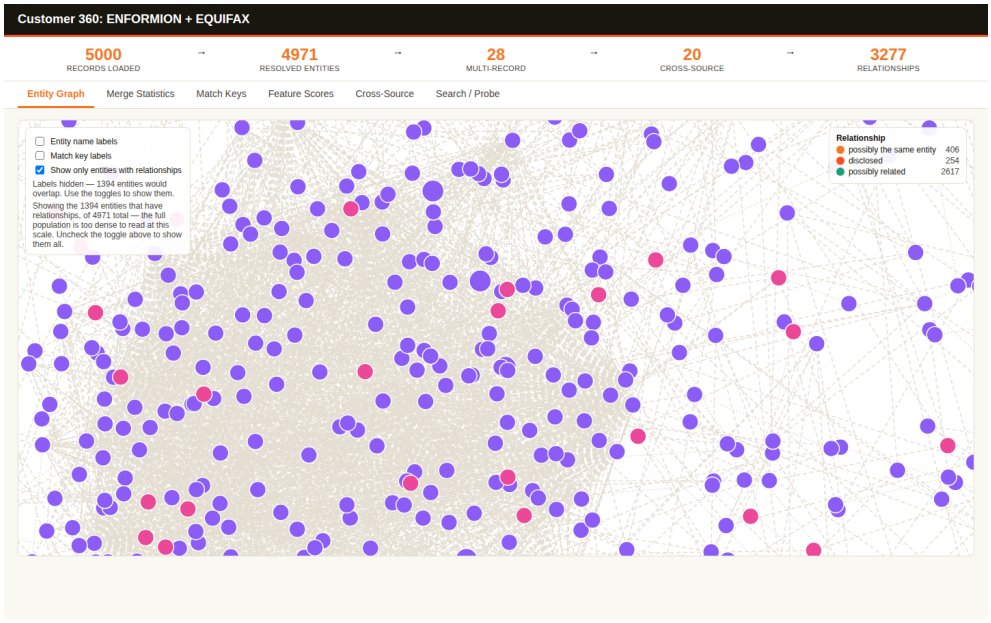
## Questions & Responses

- **Q:** What do you want to find out about your data?
  - **R:** Which customers show up in both Enformion and Equifax, and what one customer's whole record looks like across the two.
- **Q:** Shall I build queries for this list, or would you like to drop or change any of it? (your two questions beside three requirements from the business problem, marked not asked for)
  - **R:** Keep all five.
- **Q:** The quality indicators are marginal on their raw values. What would you like to do?
  - **R:** Proceed to visualization and the Discover phase.

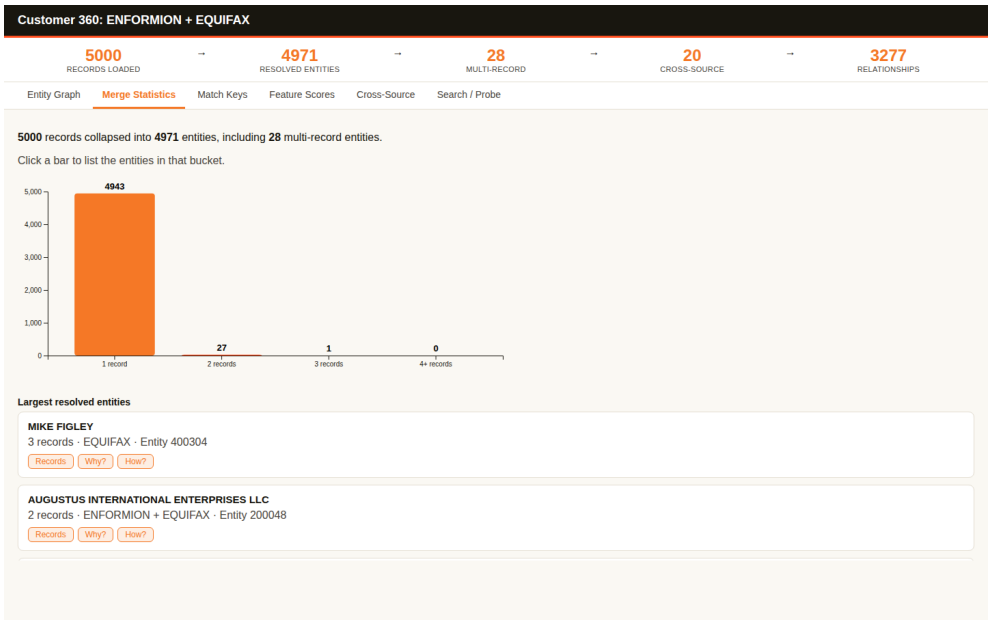
- **Q:** Would you like an interactive visualization of your resolved data?
  - **R:** Yes.
- **Q:** Would you like to explore Senzing's advanced discover capabilities using examples from your own data?
  - **R:** Yes.
- **Q:** What would you like to do next? (after Why Analysis)
  - **R:** Continue to How Analysis.
- **Q:** What would you like to do next? (after How Analysis)
  - **R:** Continue to Relationship Networks (exploring how entities connect to each other).
- **Q:** Would you like to continue to module completion now?
  - **R:** Yes.

## Actions Taken

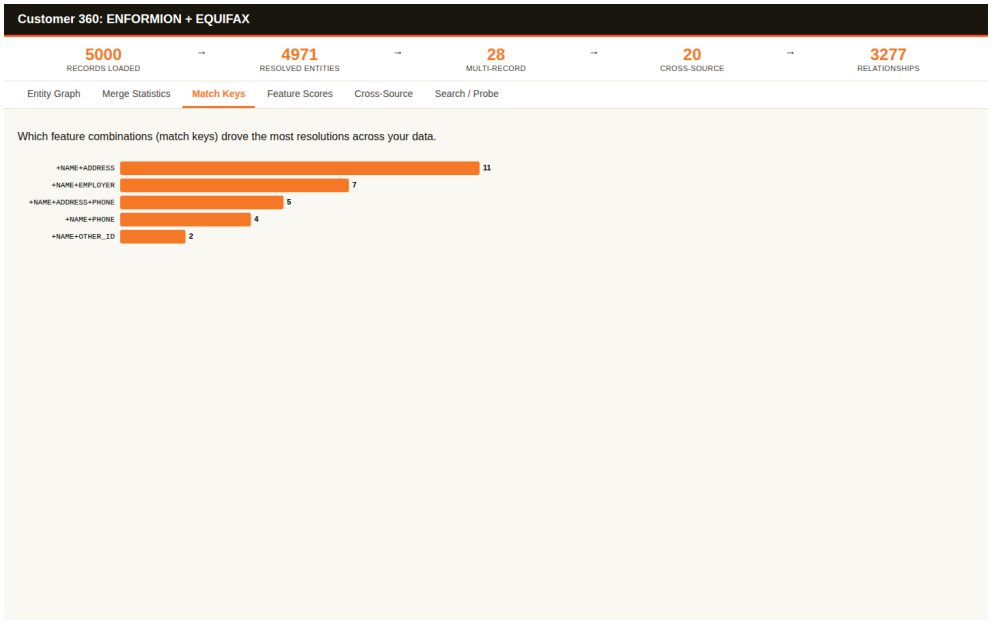
- Asked what you wanted to find out before showing anything derived, set your two questions beside three more requirements derived from the business problem's success criteria and desired output (a review queue, search, and a master-list export, each marked not asked for), and built a program for each of the five in src/query/.
- Showed the headline entity, AUGMENTI CONSULTING LLC, as a first answer once the list was confirmed, retrieved by src/query/show_entity.py through the SDK.
- Ran all five: 20 cross-source merges found, entity 360 with source provenance confirmed, a 128-pair review queue built, search verified on both an organization and a person, and a 4,971-entity master list exported to CSV and JSONL.
- Evaluated quality using proxy indicators (no truth set available), reported the marginal 8.2% possible-match rate honestly, and explained why iteration would not help - showing three specific entities as evidence rather than asserting it.
- Built the interactive visualization pointed at the bootcamper's own data, with dataset wording naming their sources rather than the Truth Set, and verified the snapshot offline with all six tabs.
- Captured and visually verified one screenshot per tab; confirmed the graph's scale-aware defaults engaged correctly at 4,971 entities (labels auto-hidden, relationship-only mode on, with an on-screen explanation).
- Ran the full Discover phase: identified candidates from 5,000 folded record keys, then demonstrated Why Analysis (entity 200048), How Analysis (entity 400304), and relationship networks with a 2-hop find_path.
- Dumped a raw find_network link element before writing the parser, which caught the MIN_ENTITY_ID/MAX_ENTITY_ID naming - under the assumed names both endpoints would have rendered blank while the match key rendered fine.
- **Found the ATLAS FOODS ambiguity:** four distinct Enformion entities sharing one name, one address, and one encrypted tax ID, unable to merge because their differing BID values map to the exclusive TRUSTED_ID feature - and therefore all four flagged ambiguous against the single Equifax record.
- Produced and verified the data-discoveries deliverable, extracting PDF text to confirm the findings are present rather than trusting the generator's success line.



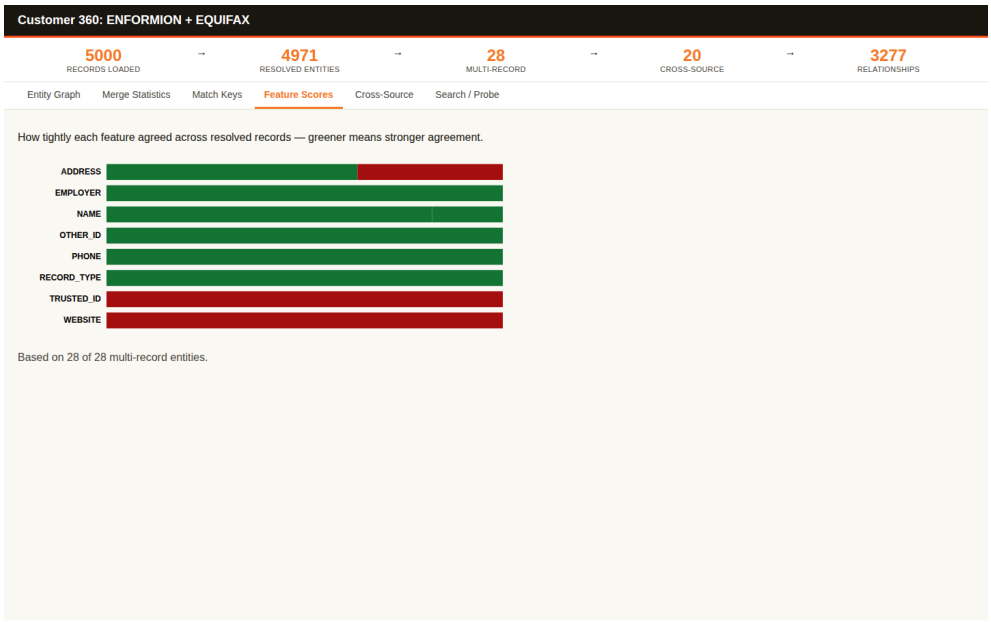

*Entity Graph - 1,394 of 4,971 entities shown in relationship-only mode with labels auto-hidden and an on-screen note explaining why, Enformion in purple and Equifax in pink, relationship legend showing 406 possibly-same, 254 disclosed and 2,617 possibly-related*



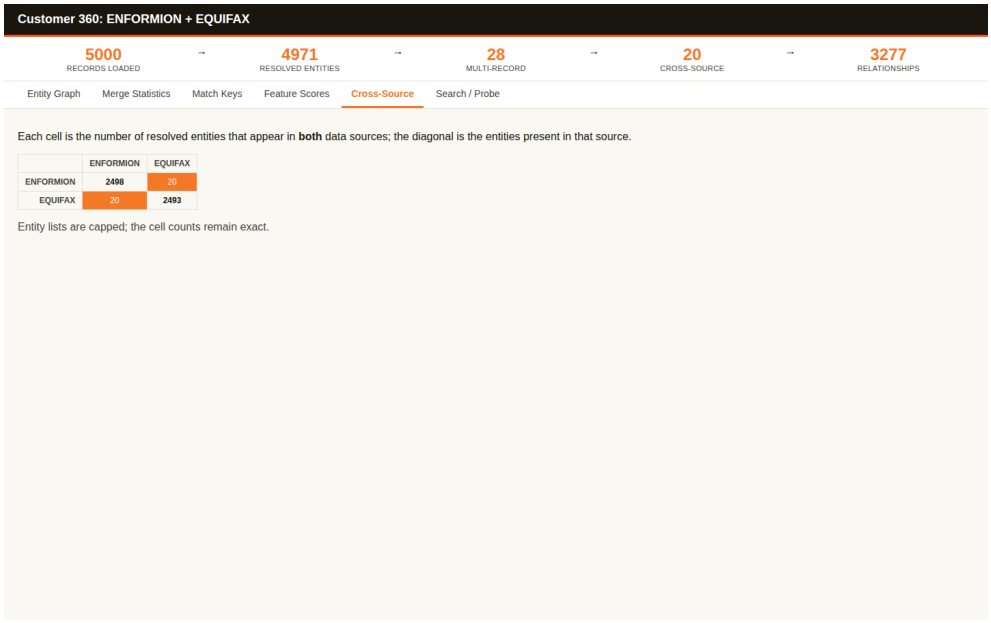

*Merge Statistics - 5,000 records collapsed into 4,971 entities, histogram dominated by 4,943 singletons with 27 pairs and 1 three-record entity, above the largest resolved entities MIKE FIGLEY and AUGUSTUS INTERNATIONAL ENTERPRISES LLC*



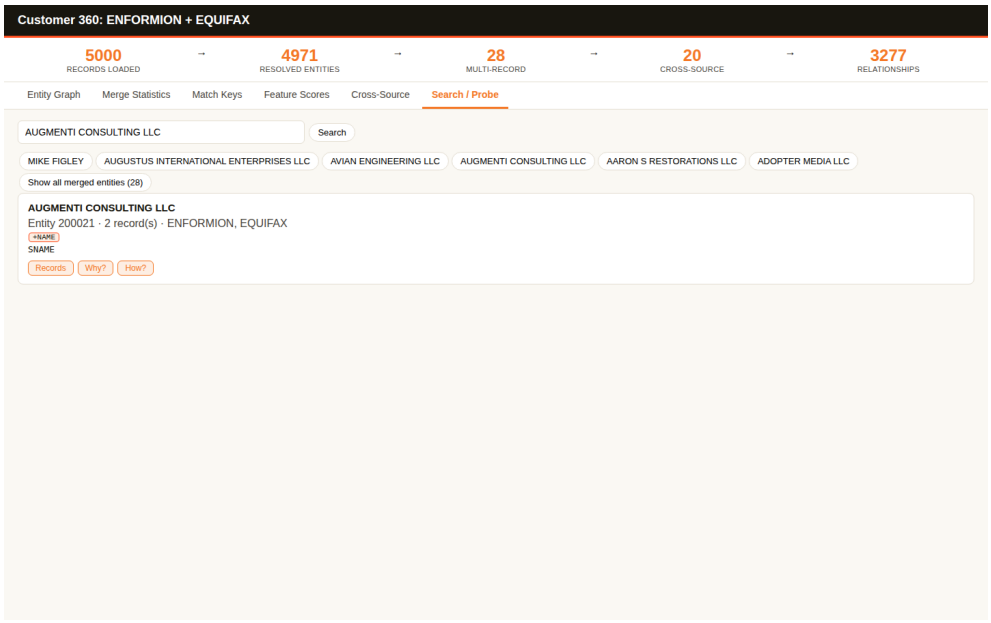

*Match Keys - only five distinct keys drove all resolutions: +NAME+ADDRESS (11), +NAME+EMPLOYER (7), +NAME+ADDRESS+PHONE (5), +NAME+PHONE (4), +NAME+OTHER_ID (2)*




*Feature Scores - per-feature agreement across all 28 multi-record entities; EMPLOYER, OTHER_ID, PHONE and RECORD_TYPE fully green, while TRUSTED_ID and WEBSITE are fully red because the two sources use different ID namespaces and website spellings for the same company*




*Cross-Source - 2x2 overlap matrix showing 20 entities shared between ENFORMION (2,498 entities) and EQUIFAX (2,493 entities)*




*Search / Probe - the query "AUGMENTI CONSULTING LLC" returning entity 200021 with 2 records spanning ENFORMION and EQUIFAX, with Records / Why? / How? actions and a "Show all merged entities (28)" browse button*

## End-of-Module Summary

**What you accomplished:**

- Built and ran five query programs that answer your business problem's success criteria directly
- Explored your own resolved data in the interactive app, and saw its defaults adapt to a dataset 60 times larger than the Truth Set
- Learned Why, How, and relationship networks on your own entities - including a 2-hop connection that exists in no single record
- Uncovered a genuine, actionable mapping question (the ATLAS FOODS ambiguity) that no aggregate statistic would have surfaced
- Left with a discoveries report whose every figure is traceable to a query you can re-run

**Files produced:**

- docs/bootcamp_data_discoveries.md and .pdf - the findings report, PDF verified by text extraction
- docs/visualizations/results_visualization.html - the offline interactive app pointed at your data
- docs/visualizations/results_visualization-*.png - six screenshots, one per tab
- docs/master_list.csv and .jsonl - the 4,971-entity master list
- docs/er_statistics.json, docs/discover_candidates.json - the underlying measurements
- src/query/ - twelve programs: sz_common.py, show_entity.py, find_cross_source_merges.py, customer_360.py, review_queue.py, search_entities.py, export_master_list.py, discover_candidates.py, why_analysis.py, why_entities.py, how_analysis.py, relationship_network.py, probe_network_shape.py

**Why it matters:**

You can now answer "who is who across these two systems?" with evidence for every claim - and just as importantly, explain precisely why 128 pairs *cannot* be answered yet, which is the difference between a result you can act on and a number you have to trust.

# Notes, Ideas and Questions

In your own words - 3 notes captured during the bootcamp

### Idea: map dba_name as a second NAME rather than payload

*2026-07-15T11:42:00-05:00 · Data Quality, Mapping, and Transformation*

The ENFORMION file has a dba_name column I mapped to payload. Trading names are how half these businesses are actually known, so a second NAME_ORG might resolve the ATLAS FOODS ambiguity without touching anything else.

**CONTEXT**

Data Quality, Mapping, and Transformation, step 4; pending question was "Ready to map the remaining columns?"

### Question: does whyEntities need a flag I have not set?

*2026-07-15T15:08:00-05:00 · Query, Visualize and Discover*

The why output does not show record-level detail. Am I missing a flag, or is that detail simply not in this response?

**ELABORATION (WRITTEN BY THE BOOTCAMP)**

The default flag set omits record data; a flags argument controls how much of the record payload the response carries.

### Reminder: check the counts against the source system before trusting them

*2026-07-15T16:20:00-05:00 · Query, Visualize and Discover*

4,971 entities from 5,241 records is a plausible number, not a verified one. Reconcile against the source system's own customer count before quoting it to anyone.

Senzing

SENZING BOOTCAMP

# Certificate of Completion

Entity Resolution, from first principles to a production pipeline

THIS CERTIFICATE IS PROUDLY PRESENTED TO

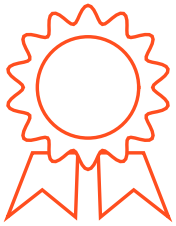
## Bootcamper

for successfully completing 9 modules of the Senzing Bootcamp - Senzing's guided path from defining the business problem to a working entity-resolution pipeline:

Entity Resolution Concepts · Discover the Business Problem · SDK setup · System verification · Truth Set visualization · Data collection · Data Quality, Mapping, and Transformation · Data processing · Query, Visualize and Discover

**July 30, 2026**

DATE COMPLETED



**Senzing Bootcamp**

ISSUED BY

*Senzing Bootcamp Claude plugin v0.6.1*